GrayCo Construction Services, LLC
4007 Broadway Rd., Bartlett, TN 38135 · (901) 585-0944 · grayco-cs.com

GrayCo Water Damage Restoration

Employee Training Manual

GrayCo's water mitigation and structural drying training for everyone who works water losses. Read every section, then answer the review questions at the end. In the CRM, open Training to tick off each part and take the quiz (pass = 16 of 20). It covers what the IICRC Water Restoration Technician (WRT) class covers, including the equipment math, so it is also your study guide for that exam.

1. Safety on a water job

Do a **hazard assessment on every job**; it is the first thing you do on the initial inspection. You can't protect yourself from a hazard you haven't noticed.

- The most common injuries on water jobs are **slips, trips and falls**. Also watch for live electricity near water, sagging ceilings and overloaded circuits.
- OSHA's order: **eliminate** the hazard, then **engineer** it out (containment, air scrubbers), and **PPE is the last resort**. One size of PPE does not fit every hazard.
- **Sewage (Category 3)**: PPE must cover **every route of exposure** (skin, eyes, mouth, lungs). The health of workers and occupants comes first.
- Every chemical has an **SDS** in the truck and on the job. Your best product information is the **SDS, the label and training**. Mix at the right dilution.
- You need a **medical evaluation and a fit test** before wearing a respirator.
- Talk to your doctor about **Hepatitis B and tetanus** shots. Hep B is three shots; after a possible exposure, get treated within **24 hours**.
- **Chemically sensitive** customers and high-risk occupants (pregnant, infants, elderly, chemo, immunocompromised, recent surgery) stay out of the work area and come back on their doctor's advice. Before applying antimicrobials: PPE on, occupants and pets out.
- **Confined spaces** (crawl spaces, attics, pits): never alone; always two people on site.
- **Lead and asbestos** are regulated by federal law. Follow GrayCo's lead standard and stop for testing if you suspect either; the PM writes a change order.
- Even clean extraction water is dumped according to **local laws and regulations**.

2. Tools and equipment

- **Moisture sensor**: beeps in wet carpet and pad and finds the edge of the water. Don't use it on anything else.
- **Moisture meters** give a number. **Non-invasive** (radio frequency) meters don't damage anything and are good for mapping; metal behind the surface gives false positives. **Invasive (pin)** meters read conductivity at the pin depth.

- **Thermo-hygrometer**: air temperature and relative humidity, recorded every visit.
- **IR camera**: shows **warmer and cooler**, not wet and dry. Always confirm with a meter.
- **Air movers**: most of our inventory. They move air across wet surfaces so the water evaporates.
- **Dehumidifiers**: **conventional refrigerant** (condensation; bottoms out around 55 GPP), **LGR** (low grain refrigerant; condensation with a pre-cooler, gets down to about **34 GPP**), and **desiccant** (chemical attraction; not limited by dew point; makes the driest air / lowest vapor pressure; check the power it needs).
- **AFDs / air scrubbers**: control dust, spores and contaminants, chosen by category of water.
- **Extraction tools**: truck mount, portables, weighted/sealed tools (the most important tools for speeding up drying).
- **Inner-air (cavity) drying**: pushes warm dry air under cabinets, into walls and under hardwood.
- The **IICRC S500** is the standard of care for water restoration; our process follows it.

3. The first visit (new claim)

1. **Get the work authorization signed**: who pays and how. No work before it is signed.
2. **Hazard assessment**, then make sure the **source is stopped**.
3. **Set expectations**: tell the customer what they will see, hear and smell, and the order of the work.
4. **Inspect** up, down and sideways to find the **edge of the water**. For each area: what is wet, how wet, is it damaged, is it salvageable? Check HVAC registers, floor outlets and crawl spaces.
5. **Protect contents**: blocks and tabs under furniture, lift drapes and clothing off the floor, raise metal items.
6. **Document**: what is wet, what is not, and **preexisting damage**, with photos. Tell the customer about preexisting damage before you start.
7. **Set the dry standard**: average several meter readings on unaffected, like materials. The **drying goal** is within about **10%** of the dry standard. Equipment stays until every material meets its goal.
8. **Extract**. Removing liquid water is the fastest way to speed drying; a quick first extraction stops the water from spreading.

Primary damage comes from the water itself. **Secondary damage** comes from the wet environment we leave behind (high humidity over 60%, mold, swelling). Our job is to stop secondary damage.

4. Categories of water (how dirty)

- **Category 1**: from a sanitary source (supply line, faucet). No substantial risk to people. It can get worse with time and dirty materials.
- **Category 2**: significant contamination that could make people sick: washer or dishwasher discharge, toilet overflow with urine, fish tanks, waterbeds.
- **Category 3**: grossly unsanitary or carrying pathogens: **sewage** from beyond the trap, and **surface or ground water** such as rising rivers. Never call it just "black water"; clean-looking water can be Category 3.
- There is **no Category 4**. Mold and regulated hazardous materials are separate special situations.
- Whether someone would drink it is **not** how you decide the category.

Category 3 rules

- Remove highly porous materials (drywall, insulation). Carpet and pad that touched Cat 3 go out. (In Cat 2, the pad goes; the carpet can be cleaned.)
- Clean every surface that stays, and apply an appropriate biocide during and after demo.
- If a customer refuses the proper Cat 3 procedures, **stop work** and call your PM. A waiver does not protect us or them.
- High-value rugs can be saved only with intensive cleaning, antimicrobial, fast drying and third-party testing.

5. Classes of loss (how much water)

Class tells you how much evaporation load there is. It is based on the **percentage of wet porous surface** in the space: ceiling + walls + floor added together.

- **Class 1**: under **5%**. Least evaporation.
- **Class 2**: **5% to 40%**. Carpet and pad wet, wicking under 24 inches up the walls.
- **Class 3**: more than **40%**. Often from overhead; walls wet above 24 inches; wet insulation. Water from above does **not** automatically make a loss Class 3 or Category 3.
- **Class 4**: specialty drying. Water is **deeply held** in low-porosity materials such as hardwood, plaster, brick, concrete, stone, or many layers. Needs **LGR or desiccant** and very dry air.

Example: a 10 x 10 x 8 room has 520 SF of surface (100 ceiling + 100 floor + 320 walls). 26 SF wet is 5%: Class 1.

6. The science of drying

Water leaves a building three ways: **extraction** (liquid, the most effective), **evaporation**, and **dehumidification or ventilation** to get the vapor out.

H.A.T., the drying pie: **Humidity, Airflow, Temperature**. Miss any one and drying stalls.

- **Relative humidity (RH)**: water in the air as a percent of what it could hold at that temperature. 100% = saturated. Warm the air and RH drops; cool it and RH rises, even with the same water in it.
- **GPP (grains per pound, humidity ratio)**: the actual amount of water in the air. GPP, **vapor pressure** and **dew point** all move together.
- **Hygroscopic** materials (wood, drywall, fabric) soak up or give off moisture depending on RH.
- **Airflow** breaks the thin layer of saturated air on a wet surface.
- **Temperature**: warming the wet material adds energy and raises its vapor pressure.
- **Vapor pressure**: moisture moves from high pressure to low pressure. Wet materials dry when the air around them has lower vapor pressure. The best combination is **warm, dry air**.
- **Dew point**: the temperature where air becomes saturated. Surfaces colder than the dew point get **condensation**. 75°F at 60% RH has a dew point of about **62°F**.

Changes of state: evaporation (liquid → vapor, energy added; the surface cools), condensation (vapor → liquid), **sublimation** (ice → vapor; freeze-drying books and documents), deposition (vapor → solid), melting, freezing.

Psychrometric chart: plug in temperature and RH to get GPP, dew point and vapor pressure. Take readings in the affected area, an unaffected area, outside, and at the dehumidifier outlet every visit.

Permeance: materials under **1 perm** (vinyl, polyethylene, vinyl wallpaper, some paints) are vapor barriers and trap water. They may have to come off.

7. Equipment math

Open, closed or combination? Compare inside and outside GPP. If outside air is wetter, use a **closed** (dehumidifiers only) system.

Dehumidifier factors

- Conventional refrigerant: Class 1 = 100, Class 2 = 40, Class 3 = 30, Class 4 = not used.
- LGR: Class 1 = 100, Class 2 = 50, Class 3 = 40, Class 4 = 40.
- Desiccant (air changes per hour): Class 1 = 1, Class 2 = 2, Class 3 = 3, Class 4 = 3.

Refrigerant (conventional and LGR): cubic feet ÷ factor = pints per day needed ÷ the unit's **AHAM rating** = number of units. **Always round up.** AHAM ratings are pints per day at 80°F and 60% RH.

- Example: Class 3, 30 x 25 x 8 = 6,000 CF ÷ 30 = 200 pints ÷ 35-pint units = 5.7 → **6**.

Desiccant: cubic feet × ACH ÷ 60 = CFM needed ÷ the unit's CFM = number of units. (The same formula sizes air scrubbers.)

- Example: Class 3, 6,000 CF × 3 ÷ 60 = 300 CFM ÷ 65 CFM units = 4.6 → **5**.

Air movers (room by room, round up):

1. **1** for each wet room.
2. **1 per 50–70 SF** of wet floor (includes the bottom 2 ft of wall).
3. **1 per 100–150 SF** of wet wall and ceiling above 2 ft.
4. **1** for each inset or offset over 18 inches.

If water only got a little way into a room (under 2 ft), use **1 per 14 linear feet** of wet wall instead. Face all air movers the same way; never let them blow against each other. In Class 4, once the surface water is gone, slow the airflow and focus on heat and dry air.

Standing water: length × width × depth in feet = cubic feet. × **7.48** = gallons. × **8.34** = pounds. Inches ÷ 12 = feet (3 inches = 0.25 ft). Pump basements slowly: gallons ÷ pump rate = time. If water keeps coming in, stop; outside soil pressure can collapse the walls.

Don't dry what you are going to tear out. Remove non-salvageable wet materials first, then set your drying plan.

8. Floors, walls and building materials

Carpet and pad

- Category 3: remove and discard both.
- Category 2: discard the pad; extract, clean and dry the carpet, then reinstall after repairs.
- Category 1: consider **in-place drying**, but only if **all** of these are yes: Category 1, under **72 hours**, deep extraction tools on hand, and LGR or desiccant available.
- Focus on how much water you **leave behind**, not how much you took out. If you can't deep-extract, float the carpet (knee kicker and awl) and pull the pad.

Other floors

- Vinyl over wet underlayment is a vapor barrier; it usually comes up.
- Carpet over hardwood: pull the carpet, discard the pad, and focus on the hardwood and subfloor.
- **Hardwood** is dry when it is within **2–4 points of its EMC** (and the subfloor within 2–4 points of the hardwood). Probe in the **dark (winter) grain**. **Cupping** (edges up, bottom wet) usually dries out. **Crowning** = top wetter. **Buckling** (lifting off the floor) usually means replacement. Tell the customer to let the floor acclimate before refinishing.
- Concrete holds up; the problem is when its moisture reaches other materials.

Wood products: plywood holds up best, then OSB; particle board is usually replaced; **MDF** suffers the most.

Walls and ceilings

- Drywall dries well and is strong again once dry, but its paper face molds if it stays wet.
- **Sagging, saturated ceilings are a safety hazard**: drain them and take them down.
- Flat paint dries easily. **Vinyl wallpaper** and gloss enamel trap water; you may need to pull the baseboard and drill holes to get air into the cavity.
- Check insulation. Wet cellulose comes out, and so does any insulation hit by Category 2 or 3 water.

Monitor at least every 24 hours: same tech, same meters, same spots. Record everything in the CRM.

9. Mold and microbes

- Mold needs **spores, food (anything organic), moisture and still air**. Take away one and it stops. On a water job, our job is to take away the water fast.
- **MVOCs** are the gases microbes give off. **Musty** = mold; **sharp or bitter** = bacteria. If you smell them, put on PPE and tell your supervisor calmly. Don't alarm the customer.
- **Sanitizer**: reduces pathogens to a safe level (the lowest level). **Disinfectant**: kills microorganisms but not necessarily spores. **Sterilizer**: kills everything, spores included.
- Use antimicrobials only when they are needed, and follow the label exactly.
- Set up containment and engineering controls **before** removing the source or turning on air movers.
- Wood moisture and decay: surface mold around **16%**, dry rot **20–29%**, wet rot **30%+**.
- Bring in a third-party **IEP** (Indoor Environmental Professional) for heavy contamination, Category 3 and high-risk occupants. Any mold, lead or asbestos found is handled by change order.

Review questions

Circle the best answer.

1. The first priority on the initial inspection is:
- A. Setting equipment
- B. A hazard assessment
- C. Moving contents
- D. Calling the adjuster

2. The most common hazards on a water job are:
- A. Slips, trips and falls
- B. Cuts and burns
- C. Chemical spills
- D. Noise

3. Before wearing a respirator you must have:
- A. A beard trim only
- B. A medical evaluation and fit test
- C. An OSHA card
- D. Nothing

4. An IR camera shows:
- A. What is wet
- B. What is warmer or cooler
- C. Moisture content
- D. Mold

5. LGR dehumidifiers can dry air down to about:
- A. 100 GPP
- B. 55 GPP
- C. 34 GPP
- D. 10 GPP

6. The drying goal is usually set within about ___ of the dry standard:
- A. 10%
- B. 25%
- C. 50%
- D. 1%

7. Damage caused by the wet environment left behind (humidity, mold) is:
- A. Primary damage
- B. Secondary damage
- C. Preexisting damage
- D. Class 4 damage

8. A toilet overflow with urine is normally:

A. Category 1
B. Category 2
C. Category 3
D. Category 4

9. Water from an overflowing river that enters a house is:

A. Category 1
B. Category 2
C. Category 3
D. Class 4

10. Carpet pad wet with Category 2 water should be:

A. Dried in place
B. Removed and discarded
C. Cleaned and reused
D. Sprayed and left

11. Wet porous materials make up 20% of the room's surfaces. The class is:

A. Class 1
B. Class 2
C. Class 3
D. Class 4

12. Water deeply held in hardwood, plaster or concrete is:

A. Class 1
B. Class 2
C. Class 3
D. Class 4

13. The three parts of the "drying pie" (H.A.T.) are:

A. Heat, air, time
B. Humidity, airflow, temperature
C. Hours, amps, temperature
D. Humidity, air movers, tools

14. Air at 75°F and 60% RH has a dew point of about:

A. 32°F
B. 50°F
C. 62°F
D. 75°F

15. A material with a permeance under 1 perm is:

A. A vapor barrier
B. Very porous
C. Class 1
D. Always salvageable

16. Class 3, 6,000 CF, conventional refrigerant (factor 30), 35-pint units. How many dehumidifiers?

A. 4
B. 5
C. 6
D. 7

17. The desiccant formula is:

A. CF ÷ factor ÷ AHAM
B. CF × ACH ÷ 60 ÷ unit CFM
C. SF ÷ 50
D. CF × 7.48

18. Air movers for wet floor: one for every:

A. 10–20 SF
B. 50–70 SF
C. 100–150 SF
D. 200 SF

19. A cubic foot of water is about how many gallons?

A. 1
B. 3.8
C. 7.48
D. 8.34

20. In-place carpet drying is only an option when it is:

A. Category 3 and under 72 hours
B. Category 1 and under 72 hours, with deep extraction and LGR/desiccant
C. Any category if the customer signs a waiver
D. Category 2 with new pad

Employee name: _____ Date: _____

Score: _____ of 20 Reviewed by: _____